

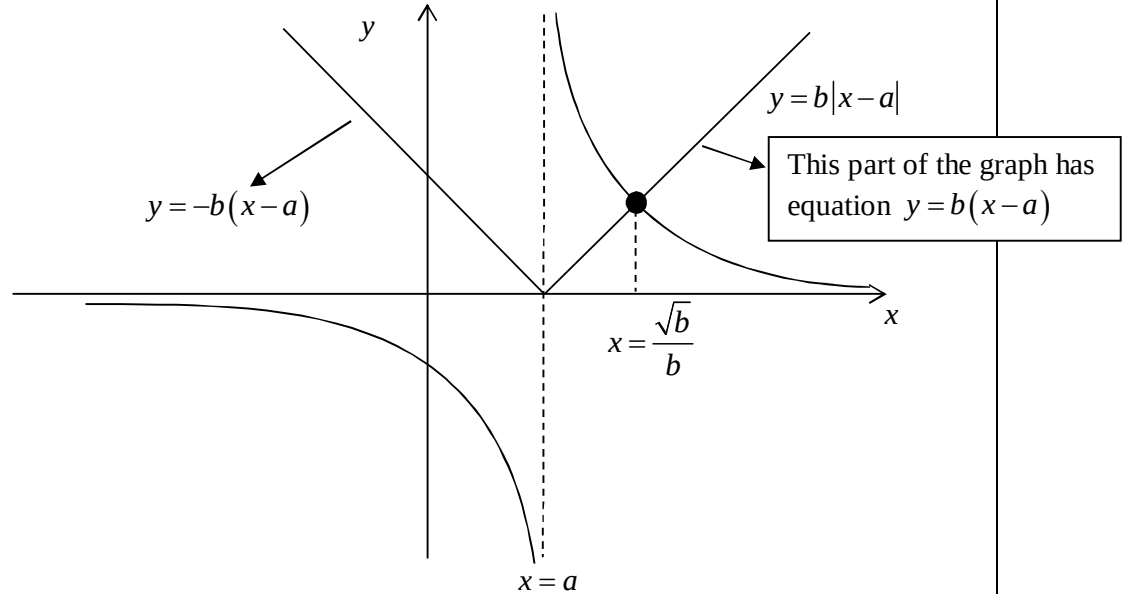
2017 A Level paper 1 Suggested Solutions

1	Suggested Solutions
	$e^{2x} \ln(1+ax) = \left(1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots\right) \left(ax - \frac{(ax)^2}{2} + \frac{(ax)^3}{3} + \dots\right)$ $= ax - \left(\frac{a^2}{2}\right)x^2 + \left(\frac{a^3}{3}\right)x^3 + 2ax^2 - 2\left(\frac{a^2}{2}\right)x^3 + \left(\frac{4a}{2!}\right)x^3 + \dots$ $= ax + \left(2a - \frac{a^2}{2}\right)x^2 + \left(2a - a^2 + \frac{a^3}{3}\right)x^3 + \dots \quad (\text{up to } x^3)$ Replace x by $2x$ in the standard series for e^x and replace x by ax in the standard series for $\ln(1+x)$ For no term in x^2, $2a - \frac{a^2}{2} = 0$ $\frac{a}{2}(4-a) = 0$ $a = 4 \quad (\text{since } a \neq 0)$ Means let the coefficient of x^2 be 0.

2	Suggested Solutions
(i)	Tips when sketching graphs with unknown constants: Given that a and b are positive constants, students are encouraged to input any positive real value of a and b into the GC to see the general shape of the graphs. Note that all axial intercepts and equation of asymptote must still be labelled with unknown constants a and b.

(ii)

To solve $\frac{1}{x-a} < b|x-a|$, students are strongly encouraged to make use of the graphs sketched in part (i)



Recall:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Therefore:

$$|x-a| = \begin{cases} x-a & \text{if } x \geq a \\ -(x-a) & \text{if } x < a \end{cases}$$

To find intersection between the 2 graphs,

$$\frac{1}{x-a} = b(x-a) \quad (\text{since } x > a)$$

$$(x-a)^2 = \frac{1}{b}$$

$$x = a + \frac{\sqrt{b}}{b} \quad \text{or} \quad x = a - \frac{\sqrt{b}}{b} \quad (\text{rej. } \because x > a)$$

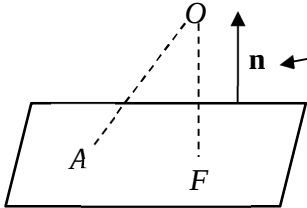
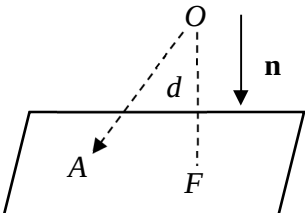
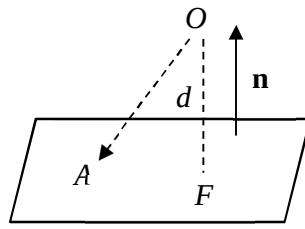
$$\therefore \frac{1}{x-a} < b|x-a|,$$

$$x < a \quad \text{or} \quad x > a + \frac{\sqrt{b}}{b}$$

3	Suggested Solutions
(i)	$y^2 - 2xy + 5x^2 - 10 = 0$ ----- (1) Differentiating with respect to x, $2y \frac{dy}{dx} - 2 \left(y + x \frac{dy}{dx} \right) + 10x = 0$ ----- (2) At stationary point, $\frac{dy}{dx} = 0$ From (2), $-2y + 10x = 0 \Rightarrow y = 5x$ Substitute $y = 5x$ into (1): $(5x)^2 - 2x(5x) + 5x^2 - 10 = 0$ $20x^2 = 10$ $x^2 = \frac{1}{2}$ $x = \frac{1}{\sqrt{2}}$ or $x = -\frac{1}{\sqrt{2}}$ $\therefore x = \frac{\sqrt{2}}{2}$ or $x = -\frac{\sqrt{2}}{2}$ The x-coordinates of the stationary points of C are $\frac{\sqrt{2}}{2}$ and $-\frac{\sqrt{2}}{2}$
(ii)	Differentiating (2) with respect to x, $2y \frac{dy}{dx} - 2 \left(y + x \frac{dy}{dx} \right) + 10x = 0$ ← $\left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} - \frac{dy}{dx} - \frac{dy}{dx} - x \frac{d^2y}{dx^2} + 5 = 0$ For the stationary point with $x > 0$, $x = \frac{\sqrt{2}}{2}$, $\frac{dy}{dx} = 0$ $\frac{d^2y}{dx^2} = \frac{-5}{4 \frac{\sqrt{2}}{2}} = \frac{-5\sqrt{2}}{4} < 0$ It is a maximum turning point when $x = \frac{\sqrt{2}}{2}$. Recall: To determine the nature of the stationary point, we can either use 1st or 2nd derivative test. In this case, students are encouraged to use the 2nd derivative test by applying implicit differentiation to differentiate equation (2) with respect to x again. Note that: At stationary point $x = \frac{\sqrt{2}}{2}$, $\frac{dy}{dx} = 0$

4	Suggested Solutions
(i)	$y = \frac{4x+9}{x+2}$ $\frac{dy}{dx} = \frac{4(x+2) - (4x+9)}{(x+2)^2}$ $\frac{dy}{dx} = -\frac{1}{(x+2)^2}$ Since $(x+2)^2 > 0$ for all $x \in \mathbb{R}$, $\frac{dy}{dx} = -\frac{1}{(x+2)^2} < 0$. Therefore, the gradient of C is negative for all points on C. Alternative method: $y = \frac{4x+9}{x+2} = 4 + \frac{1}{x+2}$ $\frac{dy}{dx} = -\frac{1}{(x+2)^2}$ Since $(x+2)^2 > 0$ for all $x \in \mathbb{R}$, $\frac{dy}{dx} = -\frac{1}{(x+2)^2} < 0$. Therefore, the gradient of C is negative for all points on C.
(ii)	$y = 4 + \frac{1}{x+2}$ The asymptotes are $y = 4$ and $x = -2$
(iii)	Note: To transform the graph of C onto the graph of $y = \frac{1}{x}$ means to start from graph C and transform to the graph of $y = \frac{1}{x}$. $y = 4 + \frac{1}{x+2} \xrightarrow{\text{replace } y \text{ by } y+4} y = \frac{1}{x+2} \xrightarrow{\text{replace } x \text{ by } x-2} y = \frac{1}{x}$ 1. Translation of 4 units in negative y direction. 2. Translation of 2 units in positive x direction. These are the two descriptions of transformations that students ought to write to gain the 2 full marks. Alternative solution: $y = 4 + \frac{1}{x+2} \xrightarrow{\text{replace } x \text{ by } x-2} y = 4 + \frac{1}{x} \xrightarrow{\text{replace } y \text{ by } y+4} y = \frac{1}{x}$ 1. Translation of 2 units in positive x direction. 1. Translation of 4 units in negative y direction These are the two descriptions of transformations that students ought to write to gain the 2 full marks.

5	Suggested Solutions
(i)	$f(x) = x^3 + ax^2 + bx + c$ $f(1) = 1 + a + b + c = 8 \Rightarrow a + b + c = 7$ $f(2) = 8 + 4a + 2b + c = 12 \Rightarrow 4a + 2b + c = 4$ $f(3) = 27 + 9a + 3b + c = 25 \Rightarrow 9a + 3b + c = -2$ Using GC, $a = -\frac{3}{2}$, $b = \frac{3}{2}$ and $c = 7$ Concept tested here: System of linear equations Recall: From the topic "Remainder and Factor Theorem, So when the polynomial $f(x) = x^3 + ax^2 + bx + c$ is divided by $(x - 1)$, the remainder is 8, it means $f(1) = 8$. Comments: It is algebraically tedious for students who have attempted to use "long division" for this part of the question.
(ii)	$f'(x) = 3x^2 - 3x + \frac{3}{2}$ $= 3\left(x - \frac{1}{2}\right)^2 + \frac{3}{4} > 0$, for all $x \in \mathbb{R}$ Therefore the gradient of the curve is always positive. Since $f'(x) > 0$, $f(x)$ is a strictly increasing function with no turning points, the curve cuts the x-axis exactly once. Therefore $f(x) = 0$ has exactly one real root. From GC, the root is -1.33 (3 s.f.). Learning Point: If the gradient of the curve is always positive, it means that the curve is strictly increasing.
(iii)	Since tangent parallel to $y = 2x - 3$, gradient is 2, hence $f'(x) = 2$ $3\left(x - \frac{1}{2}\right)^2 + \frac{3}{4} = 2$ $3\left(x - \frac{1}{2}\right)^2 - \frac{5}{4} = 0$ $\left(x - \frac{1}{2}\right)^2 = \frac{5}{12}$ $x = \frac{1}{2} \pm \sqrt{\frac{5}{12}}$ $x = \frac{1}{2} + \sqrt{\frac{5}{12}}$ or $x = \frac{1}{2} - \sqrt{\frac{5}{12}}$

6	Suggested Solutions
(i)	This is a line parallel to $\mathbf{b}$ containing the point A with position vector $\mathbf{a}$.
(ii)	This is a plane perpendicular to $\mathbf{n}$ and d units away from the origin. If d is positive, the plane is d units from origin in the direction of $\mathbf{n}$. If d is negative, the plane is d units from origin in the direction of $-\mathbf{n}$. If d is zero, the plane contains the origin.  Note that for this question: $\mathbf{n}$ is a unit vector. Explanation: In general, the equation of the plane containing the point A in scalar product form is $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$. In this case, to find the shortest distance, OF, from the origin to the plane, we use the concept of “length of projection”. $OF = \frac{ \mathbf{a} \cdot \mathbf{n} }{ \mathbf{n} } = \mathbf{a} \cdot \mathbf{n} \quad (\because \mathbf{n} \text{ is a unit vector})$ Hence for this question, to interpret geometrically $\mathbf{r} \cdot \mathbf{n} = d$, where d is a constant, we can say that d is indeed the shortest distance from the origin to the plane with perpendicular vector $\mathbf{n}$.  If d is a positive constant, it means $\mathbf{a} \cdot \mathbf{n} > 0$, angle between $\overrightarrow{OA}$ and $\mathbf{n}$ is acute and thus the plane is d units from origin in the direction of $\mathbf{n}$.  If d is a negative constant, it means $\mathbf{a} \cdot \mathbf{n} < 0$, angle between $\overrightarrow{OA}$ and $\mathbf{n}$ is obtuse and thus the plane is d units from origin in the direction of $-\mathbf{n}$.

(iii) To solve the equations $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ (line) and $\mathbf{r} \cdot \mathbf{n} = d$ (plane) is simply using the concept of solving the point of intersection between the line and plane.

$$\mathbf{r} = \mathbf{a} + t\mathbf{b} \text{ -----(1)}$$

$$\mathbf{r} \cdot \mathbf{n} = d \text{ -----(2)}$$

Substitute (1) into (2):

$$(\mathbf{a} + t\mathbf{b}) \cdot \mathbf{n} = d$$

$$\mathbf{a} \cdot \mathbf{n} + t(\mathbf{b} \cdot \mathbf{n}) = d$$

$$t(\mathbf{b} \cdot \mathbf{n}) = d - \mathbf{a} \cdot \mathbf{n}$$

$$t = \frac{d - \mathbf{a} \cdot \mathbf{n}}{\mathbf{b} \cdot \mathbf{n}}$$

Perform expansion here, make sure the "dot product" is included during the expansion.

(since $\mathbf{b} \cdot \mathbf{n} \neq 0$)

Our aim here is to solve for the parameter t .

Substitute $t = \frac{d - \mathbf{a} \cdot \mathbf{n}}{\mathbf{b} \cdot \mathbf{n}}$ into (1) to find the point of intersection.

$$\mathbf{r} = \mathbf{a} + \left(\frac{d - \mathbf{a} \cdot \mathbf{n}}{\mathbf{b} \cdot \mathbf{n}} \right) \mathbf{b}$$

The solution is the position vector of the point of intersection between the line and the plane.

7	Suggested Solutions
(i)	$\begin{aligned} \int \sin 2mx \sin 2nx \, dx &= -\frac{1}{2} \int \cos(2mx + 2nx) - \cos(2mx - 2nx) \, dx \\ &= -\frac{1}{2} \int \cos((2m + 2n)x) - \cos((2m - 2n)x) \, dx \\ &= -\frac{1}{2} \left(\frac{\sin((2m + 2n)x)}{2m + 2n} - \frac{\sin((2m - 2n)x)}{2m - 2n} \right) + C, \quad C \in \mathbb{R} \\ &= \frac{1}{4} \left(\frac{\sin((2m - 2n)x)}{m - n} - \frac{\sin((2m + 2n)x)}{m + n} \right) + C \end{aligned}$ Observe that for integration of a product of 2 trigonometric expression (either sine or cosine) with different angles ($2mx$ and $2nx$), the preferred method is to use the factor formula available from MF26. **It is algebraically tedious for students who have attempted this question using integration by parts. $\sin P + \sin Q \equiv 2 \sin \frac{1}{2}(P + Q) \cos \frac{1}{2}(P - Q)$ $\sin P - \sin Q \equiv 2 \cos \frac{1}{2}(P + Q) \sin \frac{1}{2}(P - Q)$ $\cos P + \cos Q \equiv 2 \cos \frac{1}{2}(P + Q) \cos \frac{1}{2}(P - Q)$ $\cos P - \cos Q \equiv -2 \sin \frac{1}{2}(P + Q) \sin \frac{1}{2}(P - Q)$ Factor Formula from MF26: We will use the fourth formula. An illustration on how to use the fourth formula: e.g. $\begin{aligned} \int \sin 5x \sin x \, dx &= -\frac{1}{2} \int \cos(5x + x) - \cos(4x - x) \, dx \\ &= -\frac{1}{2} \int \cos 6x - \cos 3x \, dx \end{aligned}$

(ii)	$\int_0^{\pi} (f(x))^2 dx$ $= \int_0^{\pi} (\sin 2mx)^2 + 2 \sin 2mx \sin 2nx + (\sin 2nx)^2 dx$ $= \int_0^{\pi} (\sin^2 2mx) + 2 \sin 2mx \sin 2nx + (\sin^2 2nx) dx$ $= \int_0^{\pi} \frac{1 - \cos 4mx}{2} + 2 \sin 2mx \sin 2nx + \frac{1 - \cos 4nx}{2} dx$ $= \int_0^{\pi} 1 - \frac{1}{2} \cos 4mx - \frac{1}{2} \cos 4nx + 2 \sin 2mx \sin 2nx dx$ $= \left[x - \frac{1}{8m} \sin 4mx - \frac{1}{8n} \sin 4nx + \frac{1}{2} \left(\frac{\sin((2m-2n)x)}{m-n} - \frac{\sin((2m+2n)x)}{m+n} \right) \right]_0^{\pi}$ $= \left[\pi - \frac{1}{8m} \sin 4m\pi - \frac{1}{8n} \sin 4n\pi + \frac{1}{2} \left(\frac{\sin((2m-2n)\pi)}{m-n} - \frac{\sin((2m+2n)\pi)}{m+n} \right) \right] - 0$ $= \pi \quad (\because \sin k\pi = 0 \text{ for all } k \in \mathbb{Z})$ Use result from part (i) Use double angle formula: $\cos 2x = 1 - 2 \sin^2 x$ $\therefore \cos 4nx = 1 - 2 \sin^2 2nx$ $\sin^2 2nx = \frac{1 - \cos 4nx}{2}$ In this question, m and n are all positive integers, therefore $(2m-2n)$ and $(2m+2n)$ are also positive integers. As a result, $\sin 4m\pi = \sin 4n\pi = \sin((2m-2n)\pi) = \sin((2m+2n)\pi) = 0$
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8	Suggested Solutions
(a)	$z^2(1-i) - 2z + (5+5i) = 0$ $z = \frac{2 \pm \sqrt{4 - 4(1-i)(5+5i)}}{2(1-i)}$ $= \frac{1 \pm \sqrt{1-5(2)}}{1-i}$ $= \frac{1+3i}{1-i} \cdot \frac{1+i}{1+i} \text{ or } \frac{1-3i}{1-i} \cdot \frac{1+i}{1+i}$ $= \frac{1+4i-3}{2} \text{ or } \frac{1-2i+3}{2}$ $z = -1+2i \text{ or } z = 2-i$ Note: Coefficients of the quadratic equation is not real, i.e. coefficient of z^2 is $(1+i)$ and the constant term $(5+5i)$ is not real. The most recommendation method to solve the quadratic equation with coefficients that are not real in this case is by using quadratic formula. To solve quadratic equation using the formula: $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(b) (i)	$w^2 = 1 - 2i - 1 = -2i$ $w^3 = w(w^2) = (1 - i)(-2i) = -2 - 2i$ $w^4 = (w^2)^2 = -4$ $w^4 + pw^3 + 39w^2 + qw + 58 = 0$ $-4 + p(-2 - 2i) - 78i + q(1 - i) + 58 = 0$ Comparing real components, $-4 - 2p + q + 58 = 0 \Rightarrow -2p + q = -54 \text{ ----- (1)}$ Comparing imaginary components, $-2p - 78 - q = 0 \Rightarrow -2p - q = 78 \text{ ----- (2)}$ Using GC to solve (1) and (2), $\therefore p = -6$ and $q = -66$. Note: Working must be shown clearly since calculator is not allowed. Cartesian form in this case means $x + iy, x, y \in \mathbb{R}$
(b) (ii)	$w^4 - 6w^3 + 39w^2 - 66w + 58 = 0$ Since the coefficients of the polynomial are all real, since $w = 1 - i$ is a root, its conjugate $w = 1 + i$ is also a root. Must write this condition before concluding $w = 1 + i$ is also a root. $w^4 - 6w^3 + 39w^2 - 66w + 58 = (w - (1 - i))(w - (1 + i))(w^2 + Bw + C) \leftarrow$ $= ((w - 1) - i)((w - 1) + i)(w^2 + Bw + C)$ $= (w^2 - 2w + 2)(w^2 + Bw + C)$ Comparing constant, $C = 29$ Comparing coefficient of w^3, $B = -4$ $\therefore w^4 - 6w^3 + 39w^2 - 66w + 58 = (w^2 - 2w + 2)(w^2 - 4w + 29)$ Product of two quadratic factors Note: if $w = 1 - i$ is a root, then $[w - (1 - i)]$ is a factor.

9	Suggested Solutions
(a) (i)	$U_n = S_n - S_{n-1}$ $= An^2 + Bn - A(n-1)^2 - B(n-1)$ $= An^2 + Bn - An^2 + 2An - A - Bn + B$ $= 2An - A + B \quad \leftarrow \text{Need to simplify answer as required by the question}$
(a) (ii)	Given $U_{10} = 48$, $20A - A + B = 48$ $19A + B = 48 \text{ ----- (1)}$ Given $U_{17} = 90$, $34A - A + B = 90$ $33A + B = 90 \text{ ----- (2)}$ Using GC to solve (1) and (2), $A = 3$ and $B = -9$
(b)	$r^2(r+1)^2 - (r-1)^2 r^2 = r^2(r^2 + 2r + 1 - (r^2 - 2r + 1))$ $= r^2(4r) = 4r^3 \quad (\text{Shown})$ $\therefore k = 4$ Alternative: $r^2(r+1)^2 - (r-1)^2 r^2 = r^2[(r+1) - (r-1)][(r+1) + (r-1)]$ $= r^2(2)(2r) = 4r^3 \quad (\text{Shown})$ $\therefore k = 4$ $\sum_{r=1}^n r^3 = \frac{1}{4} \sum_{r=1}^n (r^2(r+1)^2 - (r-1)^2 r^2)$ $= \frac{1}{4} (1^2 2^2 - 0^2 1^2$ $+ 2^2 3^2 - 1^2 2^2$ $+ 3^2 4^2 - 2^2 3^2$ $+ \dots$ $+ (n-1)^2 n^2 - (n-2)^2 (n-1)^2$ $+ n^2 (n+1)^2 - (n-1)^2 n^2)$ $= \frac{1}{4} n^2 (n+1)^2$

Detailed workings must be shown for "shown" question. Remember to state the value of k as required by the question.

(c)

For $\sum_{r=0}^{\infty} \frac{x^r}{r!}$, let $x_n = \frac{x^n}{n!}$ and $x_{n+1} = \frac{x^{n+1}}{(n+1)!}$.

$$\frac{x_{n+1}}{x_n} = \frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}}$$

$$\frac{x_{n+1}}{x_n} = \frac{x^{n+1}}{(n+1)!} \div \frac{x^n}{n!}$$

$$\frac{x_{n+1}}{x_n} = \frac{x^{n+1}}{(n+1)!} \times \frac{n!}{x^n} = \frac{x}{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x}{n+1} \right|$$

$$= 0 < 1 \quad \leftarrow$$

Recall:

$$n! = n(n-1) \dots (2)(1)$$

$$(n+1)! = (n+1)n(n-1) \dots (2)(1)$$

$$\therefore \frac{n!}{(n+1)!} = \frac{1}{n+1}$$

Taken from question:

D'Alembert's ratio test states that a series of the

form $\sum_{r=0}^{\infty} a_r$ **converges** when $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$

Hence, the series $\sum_{r=0}^{\infty} \frac{x^r}{r!}$ converges for all real values of x .

$$\sum_{r=0}^{\infty} \frac{x^r}{r!} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \dots$$

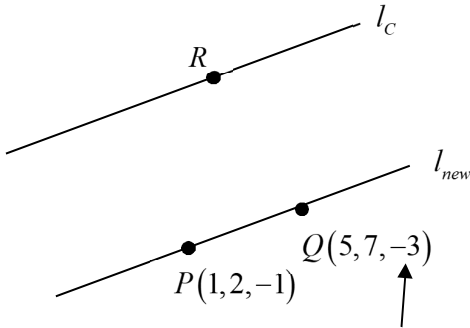
$$= 1 + x + \frac{x^2}{2} + \dots$$

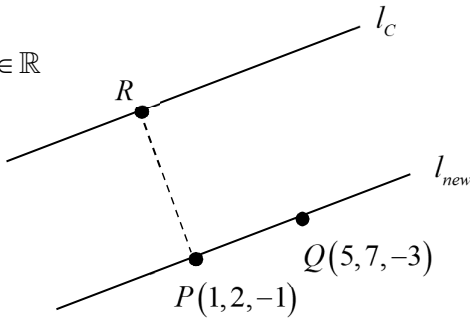
$$= e^x \quad \left. \vphantom{\sum_{r=0}^{\infty} \frac{x^r}{r!}} \right\}$$

Students are strongly encouraged to **list down the first few term** of the summation to observe the series expansion and draw links to the standard series found in MF26.

From Maclaurin expansion in MF26:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^r}{r!} + \dots$$

10	Suggested Solutions
(i)	$l_C: \mathbf{r} = \lambda \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}, \lambda \in \mathbb{R}$ $\overrightarrow{PQ} = \begin{pmatrix} 5 \\ 7 \\ a \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ a+1 \end{pmatrix}$ $l_{new}: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 5 \\ a+1 \end{pmatrix}, \mu \in \mathbb{R}$ For the cables to meet, $\begin{pmatrix} 3\lambda \\ \lambda \\ -2\lambda \end{pmatrix} = \begin{pmatrix} 1+4\mu \\ 2+5\mu \\ -1+\mu(a+1) \end{pmatrix}$, for some $\lambda, \mu \in \mathbb{R}$ $3\lambda = 1 + 4\mu$ ----- (1) $\lambda = 2 + 5\mu$ ----- (2) $-2\lambda = -1 + \mu(a+1)$ --- (3) Using GC to solve (1) and (2), $\lambda = -\frac{3}{11}$ and $\mu = -\frac{5}{11}$ Substitute $\lambda = -\frac{3}{11}$ and $\mu = -\frac{5}{11}$ into (3) $\therefore a = -\frac{22}{5}$ Cable C starts at main switching site at $(0,0,0)$, hence it must pass through $(0,0,0)$ Since R lies on l_C, $\overrightarrow{OR} = \begin{pmatrix} 3\lambda \\ \lambda \\ -2\lambda \end{pmatrix}$ for some $\lambda \in \mathbb{R}$ $\overrightarrow{RP} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 3\lambda \\ \lambda \\ -2\lambda \end{pmatrix} = \begin{pmatrix} 1-3\lambda \\ 2-\lambda \\ -1+2\lambda \end{pmatrix}$ $\overrightarrow{RQ} = \begin{pmatrix} 5 \\ 7 \\ -3 \end{pmatrix} - \begin{pmatrix} 3\lambda \\ \lambda \\ -2\lambda \end{pmatrix} = \begin{pmatrix} 5-3\lambda \\ 7-\lambda \\ -3+2\lambda \end{pmatrix}$  For this part of the question, use $a = -3$ as required by the question since the two cables do not meet.

	$\overrightarrow{RP} \cdot \overrightarrow{RQ} = \begin{pmatrix} 1-3\lambda \\ 2-\lambda \\ -1+2\lambda \end{pmatrix} \cdot \begin{pmatrix} 5-3\lambda \\ 7-\lambda \\ -3+2\lambda \end{pmatrix}$ $= \lambda^2(9+1+4) + \lambda(-18-9-8) + (5+14+3)$ $= 14\lambda^2 - 35\lambda + 22$ $= 14\left(\lambda^2 - \frac{35}{14}\lambda + \left(\frac{35}{28}\right)^2 - \left(\frac{35}{28}\right)^2\right) + 22$ $= 14\left(\lambda - \frac{35}{28}\right)^2 + \frac{1}{8}$ Since $\left(\lambda - \frac{35}{28}\right)^2 \geq 0$ for all $\lambda \in \mathbb{R}$, $14\left(\lambda - \frac{35}{28}\right)^2 + \frac{1}{8} > 0$. Since $\overrightarrow{RP} \cdot \overrightarrow{RQ} \neq 0$, it is not possible that $\angle PRQ = 90^\circ$.	
(iii)	Since R lies on l_c, $\overrightarrow{OR} = \begin{pmatrix} 3\lambda \\ \lambda \\ -2\lambda \end{pmatrix}$ for some $\lambda \in \mathbb{R}$ $\overrightarrow{PR} = \begin{pmatrix} 3\lambda \\ \lambda \\ -2\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3\lambda-1 \\ \lambda-2 \\ 1-2\lambda \end{pmatrix}$  In order for the length of PR to be as small as possible, $\overrightarrow{PR}$ has to be perpendicular to both l_c and l_{new}. $\overrightarrow{PR} \cdot \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} = 0$ $\begin{pmatrix} 3\lambda-1 \\ \lambda-2 \\ 1-2\lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} = 0$ $9\lambda - 3 + \lambda - 2 - 2 + 4\lambda = 0$ $14\lambda = 7$ $\lambda = \frac{1}{2}$ $\therefore \overrightarrow{OR} = \frac{1}{2} \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}, \text{ coordinates of } R \text{ is } \left(\frac{3}{2}, \frac{1}{2}, -1\right)$	

In order to show that angle PRQ is not 90° , we need to show that the $\overrightarrow{RP} \cdot \overrightarrow{RQ} \neq 0$ using algebraic method.

Alternative, we can show that the discriminant of $14\lambda^2 - 35\lambda + 22$ is $14\lambda^2 - 35\lambda + 22 = (-35)^2 - 4(14)(22)$
 $= -7 < 0$
 $\therefore 14\lambda^2 - 35\lambda + 22 \neq 0$

	$\overrightarrow{PR} = \begin{pmatrix} \frac{3}{2} - 1 \\ 2 \\ 1 - 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 2 \\ 0 \end{pmatrix}$ $\text{Exact Minimum length} = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{3}{2}\right)^2}$ $= \frac{\sqrt{10}}{2}$
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Qn 11	Suggested Solutions
(i)	$\frac{dv}{dt} = c, c \in \mathbb{R}$
(a)	
(i)	$\frac{dv}{dt} = c$
(b)	$v = \int c \, dt$ $v = ct + d$, where $d \in \mathbb{R}$ When $t = 0$, $v = 4$: $d = 4$ When $t = 2.5$, $v = 29$: $29 = 2.5c + 4 \Rightarrow c = 10$ $\therefore v = 10t + 4$
(ii)	$\frac{dv}{dt} = 10 - kv, k > 0$ The new rate of change of v: $\frac{dv}{dt}$ difference between the value of c which is 10 and “proportional to the velocity v, with a constant of proportionality k” which is kv $\int \frac{1}{10 - kv} \, dv = \int 1 \, dt$ Use variable separable here. $-\frac{1}{k} \ln(10 - kv) = t + a$, since $10 - kv > 0$ and $a \in \mathbb{R}$ $\ln(10 - kv) = -kt - ka$ $10 - kv = e^{-kt - ka}$ $kv = 10 - e^{-kt} \cdot e^{-ka}$ $v = \frac{10}{k} - \frac{e^{-kt} \cdot e^{-ka}}{k}$ $v = \frac{10}{k} + Ae^{-kt}$, where $A = -\frac{e^{-ka}}{k}$ Both a and k are real constants $\text{When } t = 0, v = 0: A = -\frac{10}{k}$ $\therefore v = \frac{10}{k}(1 - e^{-kt})$ Students who chose to remove modulus here must give reason.

(iii)

As $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$, $v \rightarrow \frac{10}{k}$

From the question, 'terminal velocity' is the value approached by the velocity after a long time. As shown above, when $t \rightarrow \infty$, $v \rightarrow \frac{10}{k}$ means that the terminal is $\frac{10}{k}$, where k is a real constant.

Since the terminal velocity is 40 ms^{-1} , $k = \frac{10}{40} = \frac{1}{4}$

To reach 90% of its terminal velocity, $v = \frac{90}{100} \times 40 = 36$.

Substitute $v = 36$ and $k = \frac{1}{4}$ into $v = \frac{10}{k}(1 - e^{-kt})$ to solve for t .

$$36 = 40 \left(1 - e^{-\frac{1}{4}t} \right)$$

Using GC,
 $t = 9.21034$
 $= 9.21 \text{ (3 s.f.)}$

Since answer need not be in exact form, students are encouraged to use GC (graphical method to solve for the value of t),

